HEI Haute Ecole d'Ingénierie, HES-SO Valais
Campus Energypolis
1950 Sion
Tél. 058 606 85 23

Analyse 3 - Série 25

Classe: SEf

1. Exprimer $\operatorname{artanh}(x)$ à l'aide d'un logarithme.

2. Calculer les transformées de Laplace des fonctions suivantes:

   (a) $f(t) = e^{2t}(3\sin(4t) - 4\cos(4t))$
   (b) $f(t) = (5e^{2t} - 3)^2$
   (c) $f(t) = (1 + te^{-t})^3$
   (d) $f(t) = \begin{cases} (t-1)^2 & (t > 1) \\ 0 & (0 \leq t \leq 1) \end{cases}$
   (e) $e^{-t}f(3t)$ si $\tilde{f}(s) = \dfrac{e^{-1/s}}{s}$
   (f) $f(t) = (\sin t - \cos t)^2$
   (g) $f(t) = 4\cos^2(2t)$
   (h) $f(t) = t(3\sin(2t) - 2\cos(2t))$
   (i) $f(t) = (t^2 - 3t + 2)\sin(3t)$
   (j) $f(t) = \cos(\omega t + \varphi)$
   (k) $f(t) = \sin(\omega t + \varphi)$

3. Calculer les transformées de Laplace inverses des fonctions suivantes:

   (a) $\tilde{f}(s) = \dfrac{1}{s^2 + 9}$
   (b) $\tilde{f}(s) = \dfrac{1}{s^4}$
   (c) $\tilde{f}(s) = \dfrac{1}{s^2 - 3}$
   (d) $\tilde{f}(s) = \dfrac{6}{2s - 3} - \dfrac{3 + 4s}{9s^2 - 16} + \dfrac{8 - 6s}{16s^2 + 9}$
   (e) $\tilde{f}(s) = \dfrac{6s - 4}{s^2 - 4s + 20}$
   (f) $\tilde{f}(s) = \dfrac{4s + 12}{s^2 + 8s + 16}$
   (g) $\tilde{f}(s) = \dfrac{e^{-5s}}{(s-2)^4}$
   (h) $\tilde{f}(s) = \dfrac{1}{s^2(s^2 + 1)}$
   (i) $\tilde{f}(s) = \dfrac{1}{s^3(s^2 + 1)}$
   (j) $\tilde{f}(s) = \dfrac{3s + 1}{(s-1)(s^2 + 1)}$
   (k) $\tilde{f}(s) = \dfrac{s}{(s^2 + a^2)^2}$
   (l) $\tilde{f}(s) = \ln\left(1 + \dfrac{1}{s^2}\right)$

HEI Haute Ecole d'Ingénierie, HES-SO Valais
Campus Energypolis
1950 Sion
Tél. 058 606 85 23

Analyse 3 - Série 25 (Solutions)

Classe: SEf

1. $y = \operatorname{artanh}(x) \Longleftrightarrow x = \tanh(y) = \dfrac{e^y - e^{-y}}{e^y + e^{-y}}$. Il en suit $\operatorname{artanh}(x) = \dfrac{1}{2}\ln\left(\dfrac{1+x}{1-x}\right)$.

2. (a) $\tilde{f}(s) = \dfrac{20 - 4s}{s^2 - 4s + 20}$

   (b) $\tilde{f}(s) = \dfrac{25}{s-4} - \dfrac{30}{s-2} + \dfrac{9}{s}$

   (c) $\tilde{f}(s) = \dfrac{1}{s} + \dfrac{3}{(s+1)^2} + \dfrac{6}{(s+2)^3} + \dfrac{6}{(s+3)^4}$

   (d) $\tilde{f}(s) = \dfrac{2e^{-s}}{s^3}$

   (e) $\dfrac{e^{-3/(s+1)}}{s+1}$

   (f) $\tilde{f}(s) = \dfrac{s^2 - 2s + 4}{s(s^2+4)}$

   (g) $\tilde{f}(s) = \dfrac{2}{s} + \dfrac{2s}{s^2+16}$

   (h) $\tilde{f}(s) = \dfrac{8 + 12s - 2s^2}{(s^2+4)^2}$

   (i) $\tilde{f}(s) = 6\,\dfrac{s^4 - 3s^3 + 21s^2 - 27s + 72}{(s^2+9)^3}$

   (j) $\tilde{f}(s) = \dfrac{s\cos(\varphi) - \omega\sin(\varphi)}{s^2 + \omega^2}$

   (k) $\tilde{f}(s) = \dfrac{s\sin(\varphi) + \omega\cos(\varphi)}{s^2 + \omega^2}$

3. (a) $f(t) = \dfrac{\sin(3t)}{3}$

   (b) $f(t) = \dfrac{t^3}{6}$

   (c) $f(t) = \dfrac{\sinh(\sqrt{3}\,t)}{\sqrt{3}}$

   (d) $f(t) = 3e^{3t/2} - \dfrac{1}{4}\sinh(4t/3) - \dfrac{4}{9}\cosh(4t/3) + \dfrac{2}{3}\sin(3t/4) - \dfrac{3}{8}\cos(3t/4)$

   (e) $f(t) = 2e^{2t}(3\cos(4t) + \sin(4t))$

   (f) $f(t) = 4e^{-4t}(1-t)$

   (g) $f(t) = \begin{cases} \frac{1}{6}(t-5)^3 e^{2(t-5)} & (t \geq 5) \\ 0 & (t < 5) \end{cases}$

   (h) $f(t) = t - \sin(t)$

   (i) $f(t) = \dfrac{t^2}{2} + \cos(t) - 1$

   (j) $f(t) = 2e^t - 2\cos(t) + \sin(t)$

   (k) $f(t) = \dfrac{t\sin(at)}{2a}$

   (l) $f(t) = \dfrac{2(1 - \cos(t))}{t}$

HIW Hochschule für Ingenieurwissenschaften, HES-SO Wallis
Campus Energypolis
1950 Sitten
Tel. 058 606 85 23

Analysis 3 - Serie 25

Klasse: SEd

1. Drücken Sie $\operatorname{artanh}(x)$ mit Hilfe eines Logarithmus aus.

2. Berechnen Sie die Laplace–Transformierten der folgenden Funktionen:

   (a) $f(t) = e^{2t}(3\sin(4t) - 4\cos(4t))$

   (b) $f(t) = (5e^{2t} - 3)^2$

   (c) $f(t) = (1 + te^{-t})^3$

   (d) $f(t) = \begin{cases} (t-1)^2 & (t > 1) \\ 0 & (0 \le t \le 1) \end{cases}$

   (e) $e^{-t}f(3t)$ falls $\tilde{f}(s) = \dfrac{e^{-1/s}}{s}$

   (f) $f(t) = (\sin t - \cos t)^2$

   (g) $f(t) = 4\cos^2(2t)$

   (h) $f(t) = t(3\sin(2t) - 2\cos(2t))$

   (i) $f(t) = (t^2 - 3t + 2)\sin(3t)$

   (j) $f(t) = \cos(\omega t + \varphi)$

   (k) $f(t) = \sin(\omega t + \varphi)$

3. Berechnen Sie die inversen Laplace–Transformierten der folgenden Funktionen:

   (a) $\tilde{f}(s) = \dfrac{1}{s^2 + 9}$

   (b) $\tilde{f}(s) = \dfrac{1}{s^4}$

   (c) $\tilde{f}(s) = \dfrac{1}{s^2 - 3}$

   (d) $\tilde{f}(s) = \dfrac{6}{2s - 3} - \dfrac{3 + 4s}{9s^2 - 16} + \dfrac{8 - 6s}{16s^2 + 9}$

   (e) $\tilde{f}(s) = \dfrac{6s - 4}{s^2 - 4s + 20}$

   (f) $\tilde{f}(s) = \dfrac{4s + 12}{s^2 + 8s + 16}$

   (g) $\tilde{f}(s) = \dfrac{e^{-5s}}{(s-2)^4}$

   (h) $\tilde{f}(s) = \dfrac{1}{s^2(s^2 + 1)}$

   (i) $\tilde{f}(s) = \dfrac{1}{s^3(s^2 + 1)}$

   (j) $\tilde{f}(s) = \dfrac{3s + 1}{(s-1)(s^2 + 1)}$

   (k) $\tilde{f}(s) = \dfrac{s}{(s^2 + a^2)^2}$

   (l) $\tilde{f}(s) = \ln\left(1 + \dfrac{1}{s^2}\right)$

HIW Hochschule für Ingenieurwissenschaften, HES-SO Wallis
Campus Energypolis
1950 Sitten
Tel. 058 606 85 23

Analysis 3 - Serie 25 (Lösungen)

Klasse: SEd

1. $y = \operatorname{artanh}(x) \Longleftrightarrow x = \tanh(y) = \dfrac{e^y - e^{-y}}{e^y + e^{-y}}$. Daraus folgt $\operatorname{artanh}(x) = \dfrac{1}{2}\ln\left(\dfrac{1+x}{1-x}\right)$.

2. (a) $\tilde{f}(s) = \dfrac{20-4s}{s^2-4s+20}$

   (b) $\tilde{f}(s) = \dfrac{25}{s-4} - \dfrac{30}{s-2} + \dfrac{9}{s}$

   (c) $\tilde{f}(s) = \dfrac{1}{s} + \dfrac{3}{(s+1)^2} + \dfrac{6}{(s+2)^3} + \dfrac{6}{(s+3)^4}$

   (d) $\tilde{f}(s) = \dfrac{2e^{-s}}{s^3}$

   (e) $\dfrac{e^{-3/(s+1)}}{s+1}$

   (f) $\tilde{f}(s) = \dfrac{s^2-2s+4}{s(s^2+4)}$

   (g) $\tilde{f}(s) = \dfrac{2}{s} + \dfrac{2s}{s^2+16}$

   (h) $\tilde{f}(s) = \dfrac{8+12s-2s^2}{(s^2+4)^2}$

   (i) $\tilde{f}(s) = 6\,\dfrac{s^4-3s^3+21s^2-27s+72}{(s^2+9)^3}$

   (j) $\tilde{f}(s) = \dfrac{s\cos(\varphi) - \omega\sin(\varphi)}{s^2+\omega^2}$

   (k) $\tilde{f}(s) = \dfrac{s\sin(\varphi) + \omega\cos(\varphi)}{s^2+\omega^2}$

3. (a) $f(t) = \dfrac{\sin(3t)}{3}$

   (b) $f(t) = \dfrac{t^3}{6}$

   (c) $f(t) = \dfrac{\sinh(\sqrt{3}\,t)}{\sqrt{3}}$

   (d) $f(t) = 3e^{3t/2} - \dfrac{1}{4}\sinh(4t/3) - \dfrac{4}{9}\cosh(4t/3) + \dfrac{2}{3}\sin(3t/4) - \dfrac{3}{8}\cos(3t/4)$

   (e) $f(t) = 2e^{2t}(3\cos(4t) + \sin(4t))$

   (f) $f(t) = 4e^{-4t}(1-t)$

   (g) $f(t) = \begin{cases} \frac{1}{6}(t-5)^3 e^{2(t-5)} & (t \geq 5) \\ 0 & (t < 5) \end{cases}$

   (h) $f(t) = t - \sin(t)$

   (i) $f(t) = \dfrac{t^2}{2} + \cos(t) - 1$

   (j) $f(t) = 2e^t - 2\cos(t) + \sin(t)$

   (k) $f(t) = \dfrac{t\sin(at)}{2a}$

   (l) $f(t) = \dfrac{2(1-\cos(t))}{t}$